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BICOLUNIVERSITY
POLANGUI
Polangui, Albay

Name: Kehn Rebadavia **Subject:** System Administration and Maintenance

Course, Year & Section: BSIT 4B **Prof:** Guillermo V. Red

Week 6 Laboratory Guide
Network Configuration and Management

Part 1 – DNS Configuration (BIND9)

1. **BIND9 Installation & Status (**sudo systemctl status bind9**):**

```
Installing:
  bind9  bind9-utils

Installing dependencies:
  dns-root-data

Suggested packages:
  bind9-doc

Summary:
  Upgrading: 0, Installing: 3, Removing: 0, Not Upgrading: 67
  Download size: 452 kB
  Space needed: 1,707 kB / 10.5 GB available

Get:1 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 bind9-utils amd64 1:9.20.24-1ubuntu0.3 [186 kB]
Get:2 http://ph.archive.ubuntu.com/ubuntu resolute/main amd64 dns-root-data all 2025080400build1 [6,022 B]
Get:3 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 bind9 amd64 1:9.20.24-1ubuntu0.3 [259 kB]
Fetched 452 kB in 8s (58.7 kB/s)
Selecting previously unselected package bind9-utils.
(Reading database… 136625 files and directories currently installed.)
Preparing to unpack …/bind9-utils_1%3a9.20.24-1ubuntu0.3_amd64.deb…
Unpacking bind9-utils (1:9.20.24-1ubuntu0.3)…
Selecting previously unselected package dns-root-data.
Preparing to unpack …/dns-root-data_2025080400build1_all.deb…
```

2. **Local Zone Configuration (**/etc/bind/named.conf.local**):**

```
zone "company.local"{
     type master; file "/etc/bind/db.company.local";};
```

3. **Zone Database File (**/etc/bind/db.company.local**):**

```
@ IN NS ns1.company.local. admin.company.local. (
        1       ; Serial
        604800  ; Refresh
        86400   ; Retry
        2419200 ; Expire
        6047800 ) ; Negative Cache TTL

@ IN NS ns1.company.local.
ns1 IN A 192.168.1.150
www IN A 192.168.1.150
```

4. DNS Resolution Test (nslookup www.company.local 127.0.0.1**):**

```
kehn@kehn:~$ sudo named-checkzone company.local /etc/bind/db.company.loc
zone company.local/IN: loaded serial 1
OK
kehn@kehn:~$ sudo systemctl restart bind9
kehn@kehn:~$ nslookup www.company.local 127.0.0.1
Server:         127.0.0.1
Address:        127.0.0.1#53

Name:   www.company.local
Address: 192.168.1.150

kehn@kehn:~$ _
```

Part 2 – DHCP Configuration (isc-dhcp-server**)**

1. DHCP Installation & Status:

```
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-14          linux-image-unsigned-7.0.0-14-generic    linux-modules-7.0.0-14-generic  linux-tools-7.0.0-14-generic
  linux-headers-7.0.0-14-generic  linux-main-modules-zfs-7.0.0-14-generic  linux-tools-7.0.0-14
Use 'sudo apt autoremove' to remove them.

Installing:
  isc-dhcp-server

Installing dependencies:
  isc-dhcp-common

Suggested packages:
  isc-dhcp-server-ldap  policycoreutils

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 67
  Download size: 1,281 kB
  Space needed: 4,281 kB / 10.2 GB available

Get:1 http://ph.archive.ubuntu.com/ubuntu resolute/universe amd64 isc-dhcp-server amd64 4.4.3-P1-4ubuntu2 [1,236 kB]
Get:2 http://ph.archive.ubuntu.com/ubuntu resolute/universe amd64 isc-dhcp-common amd64 4.4.3-P1-4ubuntu2 [45.8 kB]
Fetched 1,281 kB in 40s (31.7 kB/s)
Preconfiguring packages ...
Selecting previously unselected package isc-dhcp-server.
(Reading database… 139957 files and directories currently installed.)
Preparing to unpack …/isc-dhcp-server_4.4.3-P1-4ubuntu2_amd64.deb…
Unpacking isc-dhcp-server (4.4.3-P1-4ubuntu2)…
Selecting previously unselected package isc-dhcp-common.
Preparing to unpack …/isc-dhcp-common_4.4.3-P1-4ubuntu2_amd64.deb…
Unpacking isc-dhcp-common (4.4.3-P1-4ubuntu2)…
Setting up isc-dhcp-server (4.4.3-P1-4ubuntu2)…
Generating /etc/default/isc-dhcp-server...
Created symlink '/etc/systemd/system/multi-user.target.wants/isc-dhcp-server.service' → '/usr/lib/systemd/system/isc-dhcp-server.service'
Created symlink '/etc/systemd/system/multi-user.target.wants/isc-dhcp-server6.service' → '/usr/lib/systemd/system/isc-dhcp-server6.servic
Setting up isc-dhcp-common (4.4.3-P1-4ubuntu2)…
Processing triggers for man-db (2.13.1-1build1)…
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.
```

2. DHCP Subnet & Range Configuration (/etc/dhcp/dhcpd.conf**):**

```
#   }
#}
  subnet 192.168.0 netmask 255.255.255.0 {
      range 192.168.1.160 192.168.1.170;
      option routers 192.168.1.1;
      option domain-name-servers 8.8.8.8, 8.8.8.4;
}



kehn@kehn:~$
kehn@kehn:~$ sudo restart
```

Part 3 – HTTP Service (Apache2**)**

1. Apache Installation & Service Status:

```
Setting up libaprutil1-ldap:amd64 (1.6.3-3ubuntu3.1)…
Setting up libaprutil1-dbd-sqlite3:amd64 (1.6.3-3ubuntu3.1)…
Setting up apache2-utils (2.4.66-2ubuntu2.5)…
Setting up apache2-bin (2.4.66-2ubuntu2.5)…
Setting up apache2 (2.4.66-2ubuntu2.5)…
Enabling module mpm_event.
Enabling module authz_core.
Enabling module authz_host.
Enabling module authn_core.
Enabling module auth_basic.
Enabling module access_compat.
Enabling module authn_file.
Enabling module authz_user.
Enabling module alias.
Enabling module dir.
Enabling module autoindex.
Enabling module env.
Enabling module mime.
Enabling module negotiation.
Enabling module setenvif.
Enabling module filter.
Enabling module deflate.
Enabling module status.
Enabling module reqtimeout.
Enabling conf charset.
Enabling conf localized-error-pages.
Enabling conf other-vhosts-access-log.
Enabling conf security.
Enabling conf serve-cgi-bin.
Enabling site 000-default.
Created symlink '/etc/systemd/system/multi-user.target.wants/apache2.service' → '/usr/lib/systemd/syst
Created symlink '/etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service' → '/usr/lib/
Processing triggers for ufw (0.36.2-9build1)…
Processing triggers for man-db (2.13.1-1build1)…
Processing triggers for libc-bin (2.43-2ubuntu2.4)…
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.
```

- **2. Browser Test (**http://192.168.1.150**):**

```
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo
    link/ether 08:00:27:87:f3:e0 brd ff:ff:ff:ff:ff:ff
    altname enx08002787f3e0
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dyna
       valid_lft 80668sec preferred_lft 80668sec
    inet6 fd00::a00:27ff:fe87:f3e0/64 scope global dynamic mngtm
       valid_lft 86240sec preferred_lft 14240sec
    inet6 fe80::a00:27ff:fe87:f3e0/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
kehn@kehn:~$ curl http://127.0.0.1

echo "Welcome to Company Intranet Server IP: 192.168.1.150"
kehn@kehn:~$ _
```

Part 4 – FTP Service (vsftpd**)**

1. FTP Configuration (/etc/vsftpd.conf**):**

```
Use 'sudo apt autoremove' to remove them.

Installing:
  vsftpd

Installing dependencies:
  libwrap0  ssl-cert

Summary:
  Upgrading: 0, Installing: 3, Removing: 0, Not Upgrading: 67
  Download size: 189 kB
  Space needed: 495 kB / 10.2 GB available

Get:1 http://ph.archive.ubuntu.com/ubuntu resolute/main amd64 libwrap0 amd64 7.6.q-36build2 [48.5 kB]
Get:2 http://ph.archive.ubuntu.com/ubuntu resolute/main amd64 ssl-cert all 1.1.3ubuntu2 [18.8 kB]
Get:3 http://ph.archive.ubuntu.com/ubuntu resolute/main amd64 vsftpd amd64 3.0.5-0.4 [122 kB]
Fetched 189 kB in 2s (106 kB/s)
Preconfiguring packages ...
Selecting previously unselected package libwrap0:amd64.
(Reading database… 139991 files and directories currently installed.)
Preparing to unpack …/libwrap0_7.6.q-36build2_amd64.deb…
Unpacking libwrap0:amd64 (7.6.q-36build2)…
Selecting previously unselected package ssl-cert.
Preparing to unpack …/ssl-cert_1.1.3ubuntu2_all.deb…
Unpacking ssl-cert (1.1.3ubuntu2)…
Selecting previously unselected package vsftpd.
Preparing to unpack …/vsftpd_3.0.5-0.4_amd64.deb…
Unpacking vsftpd (3.0.5-0.4)…
Setting up libwrap0:amd64 (7.6.q-36build2)…
Setting up ssl-cert (1.1.3ubuntu2)…
Created symlink '/etc/systemd/system/multi-user.target.wants/ssl-cert.service' → '/usr/lib/systemd/system/ssl-cert.service'.
Setting up vsftpd (3.0.5-0.4)…
Created symlink '/etc/systemd/system/multi-user.target.wants/vsftpd.service' → '/usr/lib/systemd/system/vsftpd.service'.
/usr/lib/tmpfiles.d/vsftpd.conf:1: Line references path below legacy directory /var/run/, updating /var/run/vsftpd/empty → /run/vsftpd
tmpfiles.d/ drop-in file accordingly.
Processing triggers for man-db (2.13.1-1build1)…
Processing triggers for libc-bin (2.43-2ubuntu2.4)…
Scanning processes...
```

2. FTP Local Connection Test (ftp 192.168.1.150**):**

```
# daemon started from an initscript.
listen=NO
#
# This directive enables listening on IPv6 sockets. By default, listening
# on the IPv6 "any" address (::) will accept connections from both IPv6
# and IPv4 clients. It is not necessary to listen on *both* IPv4 and IPv6
# sockets. If you want that (perhaps because you want to listen on specific
# addresses) then you must run two copies of vsftpd with two configuration
# files.
listen_ipv6=YES
#
# Allow anonymous FTP? (Disabled by default).
anonymous_enable=NO
#
# Uncomment this to allow local users to log in.
local_enable=YES
#
# Uncomment this to enable any form of FTP write command.
write_enable=YES
#
# Default umask for local users is 077. You may wish to change this to 022,
# if your users expect that (022 is used by most other ftpd's)
#local_umask=022
#
# Uncomment this to allow the anonymous FTP user to upload files. This only
# has an effect if the above global write enable is activated. Also, you will
# obviously need to create a directory writable by the FTP user.
#anon_upload_enable=YES
```

Part 5 – Security & Firewall (UFW)

1. Firewall Rules and Status (sudo ufw status verbose**):**

```
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
53/tcp                     ALLOW       Anywhere
80/tcp                     ALLOW       Anywhere
21/tcp                     ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)
53/tcp (v6)                ALLOW       Anywhere (v6)
80/tcp (v6)                ALLOW       Anywhere (v6)
21/tcp (v6)                ALLOW       Anywhere (v6)

kehn@kehn:~$
kehn@kehn:~$ curl http://192.168.1.150
```